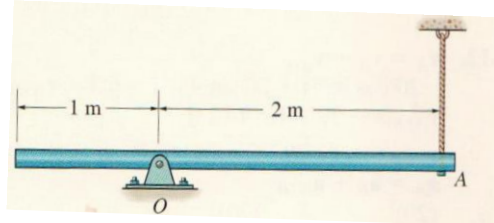


Basics. Check the following statement correct or not, and explain why.

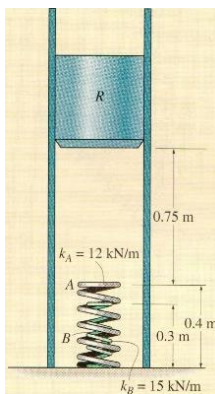
- S1. A rigid body rotates about a fixed axis x , with the angular rotation rate, ω , the kinetic energy of the body is $T = \frac{1}{2} I_x \omega^2$ where I_x is the moment of the inertia of the body about the axis, x .
- S2. A wheel subject to an external force moves along a horizontal rough surface, the friction force from ground on the wheel is $\mu_s N$. μ_s is the static friction coefficient between the wheel and rough surface, and N is the normal supporting force from ground surface to the wheel.
- S3. Two balls on a smooth horizontal surface collide to each other, the total kinetic energy of the two-balls system after collision will be reduced.
- S4. Vibration is a common phenomenon in mechanic systems. Identify the types of vibration described by the following equation
- a) $m\ddot{x} + kx = 0$
 - b) $m\ddot{x} + kx = k\delta_m \sin \omega_f t$
 - c) $m\ddot{x} + c\dot{x} + kx = 0$

Problem 1. The 20-kg slender rod is pinned at O. For the moment the cable attached at the end A suddenly breaks, you are asked to

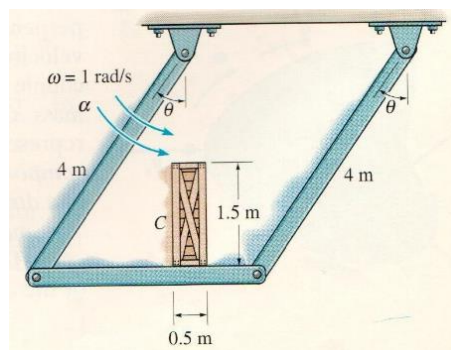
- provide forces (free body diagram) and motion analysis
- determine the acceleration of the end A
- find out the reaction at the pin support.



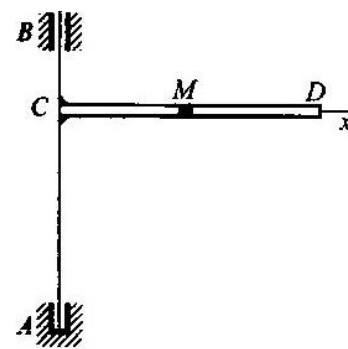
Problem 2. The ram R has a mass of 100 kg and is released from rest 0.75 m from the top of a spring, A , that has a stiffness $k_A = 12$ kN/m. If a second spring B , having a stiffness $k_B = 15$ kN/m, is “nested” in A , determine the maximum displacement of A needed to stop the downward motion of the ram. The unstretched length of each spring is indicated in the figure. Neglect the mass of the springs. You are required to provide process analysis and make drawing for each process start-end location clearly.



Problem 2



Problem 3



Problem 4

Problem 3. The 50-kg uniform crate rests on the platform for which the coefficient of static friction is $\mu_s = 0.5$. If at the instant $\theta = 30^\circ$ the supporting links have an angular velocity $\omega = 1$ rad/s and angular acceleration $\alpha = 0.5$ rad/s², determine the friction force on the crate.

Hints: (1) identify type of motion of the crate, (2) find horizontal acceleration of the crate, (3) find vertical normal supporting force on the crate from the plate.

Problem 4. A horizontal pipe CD freely rotates about an axis AB. A small ball with mass m is initially put inside the pipe at M ($MC = a$). Given the pipe an initial rotation rate of ω_0 , determine when the ball reaches the exit D (1) the small ball's rotation rate, ω , and (2) the force the pipe acting on the ball at D. Free-body diagram is required.